ANALYSIS REPORT

Wave 5.2R Offline Leader Local Promotion Gate Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Decision

K01 and H08 passed the local qualification gates required before a bounded cross-surface promotion campaign. The result authorizes campaign execution; it does not promote either candidate to global-leader status.

The accepted periodic GRU and periodic harmonic MLP remain unchanged. If the new candidates later pass the `Fw`, `Bw`, direction-aware `global`, official TE Curve Verification Pipeline, and target-runtime gates, the intended outcome is a four-leader portfolio rather than replacement of the incumbents.

Evaluated Candidates

Lane	Candidate	Architecture role	Local gate result
Temporal	K01	causal GRU residual over an analytical harmonic anchor	passed
Non-temporal	H08	condition-dependent harmonic coefficient residual	passed

K01 and H08 are Wave 5.2R physics-guided models. They are not interchangeable with the two established non-PINN deployment references.

K01 Local Evidence

The wider replay tolerance below applies only to the frozen CUDA-reference payload; strict same-runtime Python/ONNX comparisons use the tighter tolerance.

Check	Result	Interpretation
CPU replay versus saved CUDA payload, maximum	`2.8625131e-05 deg`	passed cross-device tolerance
CPU replay versus saved CUDA payload, mean	`3.8195753e-07 deg`	negligible average drift
CPU replay versus saved CUDA payload, P99	`3.2027310e-06 deg`	bounded tail drift
Reset reproducibility	`0 deg`	deterministic reset
Chunked versus full-sequence output	`7.4505806e-09 deg`	stateful chunk equivalence passed
Mutated-future prefix difference	`0 deg`	causal-prefix gate passed
ONNX curve parity	`1.4901161e-08 deg`	Python/ONNX parity passed
ONNX hidden-state parity	`5.9604645e-07`	recurrent-state parity passed
State-carry effect	`0.0013447043 deg`	carried state measurably affects output
Chunk latency P95	`271.85 us`	local `500 us` proxy passed
Chunk latency maximum	`364.4 us`	local proxy remained bounded

Check	Result	Interpretation
ONNX package size	`647963 bytes`	compact export package
Trainable parameters	`159891`	recorded deployment scale

H08 Local Evidence

Check	Result	Interpretation
Saved curve replay	`2.9802322e-08 deg`	deterministic reconstruction passed
Saved coefficient replay	`7.4505806e-09`	coefficient replay passed
Repeat difference	`0`	deterministic repeat
ONNX curve parity	`2.9802322e-08 deg`	Python/ONNX parity passed
ONNX coefficient parity	`7.4505806e-09`	inspectable coefficient parity passed
Condition latency P95	`60.705 us`	local latency proxy passed
Condition latency maximum	`197.2 us`	local proxy remained bounded
ONNX package size	`320299 bytes`	compact export package
Trainable parameters	`7651`	low-complexity non-temporal candidate

The exported H08 path keeps the coefficients, residual correction, analytical anchor, and reconstructed curve individually inspectable.

Validity And Fallback Behavior

The validity probe covered five cases. The single in-envelope valid case was accepted; four invalid or out-of-envelope cases were routed to the required fallback. The local checks did not replace the established deployment references and did not claim TwinCAT runtime acceptance.

Prepared Cross-Surface Campaign

The approved campaign contains three seeds (`314159` , `271828` , and `161803`) on three distinct surfaces:

- direction-specific `Fw` ;
- direction-specific `Bw` ;
- direction-aware `global` .

  K01 and H08 contribute `18` promotion runs. Nine matched H04 analytical-anchor runs make the full queue `27` runs. Each direction receives a training-only analytical-anchor fit; a forward anchor is never reused as a backward or global anchor. The campaign is prepared and preflight-clean. Its scalar leaderboard is bookkeeping evidence only. Promotion requires a separate official multi-index, curve-first TE Curve Verification Pipeline refresh with distinct `global` , `Fw` , and `Bw` decisions.

Promotion Boundary

K01 or H08 may join the four-leader portfolio only after all of the following:

1. repeatable cross-seed training on every required surface;
2. acceptable raw, shape, offset, continuity, harmonic, phase, and robustness evidence;
3. no disqualifying direction-specific regression;
4. official curve-first verification;
5. sufficient export and target-runtime evidence.

Until then:

- K01 is the temporal offline leader;
- H08 is the balanced non-temporal offline leader;
- periodic GRU remains the accepted temporal non-PINN reference;
- periodic harmonic MLP remains the accepted non-temporal non-PINN reference.

Integrated Specialist Model TODO

A next-step design study must evaluate whether one inspectable, PLC-conscious architecture can combine complementary mechanisms without inheriting their individual weaknesses:

- K01 temporal context, raw-error control, and offset behavior;
- H08 balanced harmonic and phase behavior;
- F01 centered-shape fidelity;
- S01 harmonic amplitude and phase specialization;
- H04 analytical interpretability and low-frequency structure;
- Stage 10 R00 sparse residual structure;
- Stage 10 S01 symbolic or compact correction structure.

This is a recorded future TODO, not authorization to start Wave 6 or a new training campaign. It requires its own technical design and approval gate.

Reproducibility Evidence

The canonical local evidence is stored under:

`output/validation_checks/wave52r_offline_leader_promotion/2026-07-30-19-24-35__wave52r_offline_leader_promotion/`

The package includes `promotion_gate_summary.yaml`, `promotion_gate_report.md`, `artifact_inventory.csv`, and the two ONNX exports.